

NORTH SOUTH UNIVERSITY

PROJECT REPORT

SUBSCRIPTION MANAGER

A Java Object-Oriented Programming Project

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Group Members Contribution

Name	ID	Contribution
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1. Introduction

The Subscription Manager is a Java application for managing monthly and annual subscriptions. It stores subscription details, calculates monthly expenses, checks budget usage, allows subscriptions to be cancelled, and generates a summary report. The project also uses basic OOP concepts such as abstraction, inheritance, polymorphism, encapsulation, and interfaces.

2. Project Objectives

- ✓ To develop a simple system for managing different types of subscriptions.
- ✓ To store and display important subscription information.
- ✓ To calculate monthly subscription expenses.
- ✓ To distinguish between monthly and annual subscription costs.
- ✓ To provide a cancellation mechanism for subscriptions.
- ✓ To monitor subscription expenses against a user-defined monthly budget.
- ✓ To generate a summary report showing total, active, cancelled subscriptions and monthly expenses.
- ✓ To demonstrate practical use of Java OOP concepts

3. Project Features

- ✓ Add Monthly Subscription
- ✓ Add Annual Subscription
- ✓ Display All Subscriptions
- ✓ Calculate Total Monthly Expenses
- ✓ Check Budget Progress
- ✓ Cancel a Subscription
- ✓ Display Cancellation Steps and Confirmation
- ✓ Generate Subscription Summary Report
- ✓ Exit the application

4. System Design and Class Structure

I. Subscription Class

Subscription is the abstract base class of the project. It contains private fields for subscription name, subscription ID, payment method, status, start date, auto-renewal, duration, and cost. Getters and setters are provided to access and modify these private fields. The class also declares the abstract `getMonthlyCost()` method and provides `displayInfo()` for displaying subscription details.

II. MonthlySubscription Class

`MonthlySubscription` extends `Subscription` and implements `Cancelable`. Its `getMonthlyCost()` method returns the subscription cost directly because the cost represents the monthly amount. It also provides cancellation and display methods.

III. AnnualSubscription Class

`AnnualSubscription` extends `Subscription` and implements `Cancelable`. Its `getMonthlyCost()` method divides the annual cost by 12 to calculate the equivalent monthly cost. It also supports cancellation and displays the calculated monthly cost.

IV. Cancelable Interface

The `Cancelable` interface defines the cancellation-related behavior used by the subscription classes. It declares `cancelSubscription()`, `isCancelable()`, and `getCancellationSteps()`.

V. BudgetProgressBar Class

`BudgetProgressBar` uses Java Swing to create a progress bar. It calculates the percentage of the monthly budget used from monthly expenses and the entered budget limit. The percentage is displayed on a graphical progress bar.

VI. Report Class

The `Report` class stores the total number of subscriptions, active subscriptions, cancelled subscriptions, and total monthly expenses. It displays these values as a summary report.

5. Object-Oriented Programming Concepts

i. Encapsulation

The fields of Subscription are declared private. Public getter and setter methods are used to access or update the data. This protects the internal data of the object and demonstrates encapsulation.

ii. Inheritance

MonthlySubscription and AnnualSubscription inherit from Subscription. This allows both classes to reuse common fields, constructors, getters, setters, and display functionality.

iii. Polymorphism

The List<Subscription> in Main can contain both MonthlySubscription and AnnualSubscription objects. When getMonthlyCost() or displayInfo() is called through a Subscription reference, the overridden method of the actual object is used.

iv. Interface

The Cancelable interface provides a common cancellation contract. Both MonthlySubscription and AnnualSubscription implement this interface and provide their own cancellation behavior.

v. Abstraction

The Subscription class is declared abstract and contains the abstract getMonthlyCost() method. This allows common subscription information to be defined in one base class while the actual monthly-cost calculation is implemented by the subclasses.

6. Conclusion

The Subscription Manager project helped us understand how Java and OOP concepts can be used to solve a real-life problem. We learned how to manage different types of subscriptions, calculate expenses, check the budget, and handle cancellation. Working on this project also gave us practical experience with abstraction, inheritance, polymorphism, encapsulation, and interfaces. Overall, it was a useful experience that improved our understanding of Java programming and teamwork.